Project Topics for IDC409

1. One should use numpy, pandas to store data.
2. Database to be use: read, write and modify.
3. Exploratory Data Analysis to be done and demonstrated.
4. Features to be understood and to be seen if one can reduce.
5. Machine learning model such as some regression, Decision Tree, SVM, BDT, NN, CNN etc. to be used
6. Evaluate the results using some metrics.
7. Use github to share the code.

1. Continuum Suppression

High Energy Physics classification: separate signal events from continuum background.

Main idea

Use Belle II detector features to separate “Signal” and “Background” events.
A strong binary-classification project using real scientific features.
One should understand features, reduce them and compare classifiers.

Data and EDA

You will be provided simulated/analysis-level Belle II MC dataset.
Store raw and cleaned data using CSV/SQLite.
EDA: feature distributions, class balance, correlations and signal/background separation plots.

AI / ML task

Baseline: simple cut or majority-class model.
Models: Logistic Regression, Decision Tree, Random Forest, SVM or Gradient Boosting.
Feature reduction: correlation, feature importance, PCA or recursive feature elimination.

Minimum deliverables

Clean notebook with data loading, EDA and model pipeline.
At least 3 model results in one comparison table.
Feature importance / reduced-feature model.
GitHub repository with README and instructions.

Evaluation and extension

Metrics: accuracy, precision, recall, F1, ROC-AUC.
Plot confusion matrix and ROC curve.
Extension: compare full-feature vs reduced-feature performance.

Suggested data / source links

1. Data: Provided Belle II sample **2. Belle II public pages:** https://www.belle2.org/

2. Event Type Classification

Multiclass HEP problem: classify uubar, ddbar, ssbar, ccbar and bbbar events.

Main idea

Classify different event types in High Energy Physics.
This is a natural multiclass classification project after binary classification.
One compare how well features separate light-quark, charm and bottom events.

Data and EDA

You will be provided Belle II MC data with labels such as uubar, ddbar, ssbar, ccbar and bbbar.
EDA: label distribution, feature histograms per class, pair plots for top variables.
Check missing values and highly correlated/redundant features.

AI / ML task

Models: Decision Tree, Random Forest, SVM, KNN, Logistic Regression.
Use train/validation/test split or cross-validation.
Try feature scaling for KNN/SVM and feature selection for simpler models.

Minimum deliverables

Clean multiclass ML pipeline.
Confusion matrix with class-wise interpretation.
Model comparison table and reduced-feature experiment.
GitHub code, environment file and README.

Evaluation and extension

Metrics: accuracy, macro-F1, weighted-F1, per-class precision/recall.
Discuss which event types are confused most often.
Extension: one-vs-rest ROC curves or probability calibration.

Suggested data / source links

1. Data: Provided Belle II sample **2. Belle II detector:** https://www.belle2.org/

3. Web Scraping I: Finance + News

Collect market/news data and build a simple price or direction-prediction model.

Main idea

Collect share prices along with gold, petrol, currency and news keywords.
Study whether market movement has any relation with recent news keywords.
Keep prediction modest: next-day direction or return range is better than exact price.

Data and EDA

Use yfinance/Alpha Vantage for share and currency data; news via RSS/API or selected sites.
Store daily records in SQLite: date, price, return, keyword counts, news sentiment.
EDA: trend plots, moving averages, return distribution and keyword frequency.

AI / ML task

Regression: predict next-day price/return.
Classification: Up/Down or High/Low movement.
Models: Linear/Logistic Regression, Decision Tree, Random Forest, SVM.

Minimum deliverables

Scraper/API collection script with clear source notes.
SQLite database and cleaned CSV export.
EDA notebook plus model notebook.
GitHub with setup instructions and sample output.

Evaluation and extension

Regression: MAE, RMSE, R^2; Classification: F1, confusion matrix.
Use time-based train/test split, not random split.
Extension: compare model with a naive “tomorrow = today” baseline.

Suggested data / source links

1. yfinance docs: https://ranaroussi.github.io/yfinance/ **2. Alpha Vantage:** https://www.alphavantage.co/ **3. News API:** https://newsapi.org/

API : Application Programming Interface

4. Web Scraping II: Faculty Profiles

Analyze academic profiles from Indian institutions to find patterns in training and expertise.

Main idea

Scrape faculty profile pages from Indian academic institutions.
Study patterns in degree, PhD institute, postdoc place, skills and current position.
Web scraping, text cleaning and categorical data analysis.

Data and EDA

Start with one or two institutions to keep scope manageable.
Fields: name, designation, department, PhD institute, research area, keywords, email optional.
EDA: department-wise counts, keyword clouds, PhD-country/institute trends.

AI / ML task

Text processing: keyword extraction, Term Frequency-Inverse Document Frequency and clustering of research areas.
Classification optional: predict broad department/research area from profile text.
Similarity search: find faculty with similar expertise.

Minimum deliverables

Ethical scraper with robots/terms awareness and rate limiting.
SQLite faculty-profile database.
EDA notebook and cleaned table.
Final summary: what patterns are visible and what limitations remain.

Evaluation and extension

For clustering: silhouette score + human interpretation.
For classification: accuracy/F1 with clear labels.
Extension: compare IISER/IIT/NIT profiles or map research keywords.

Suggested data / source links

1. IISER Mohali people: IISER Mohali website **2. Kaggle Vidwan profiles:** https://www.kaggle.com/datasets/mannacharya/vidwan-indian-faculty-and-researcher-dataset

5. Keyword Identification

Scrape website/news text and extract repeated words, important keywords and themes.

Main idea

Collect text from institute websites, funding agencies or news websites.
Find repeated words and meaningful keywords.
Compare two sources, for example IISER Mohali vs IISER Pune or News 1 vs News 2.

Data and EDA

Collect article/page title, URL, date and main text.
Clean HTML, remove stopwords, punctuation and duplicate articles/pages.
EDA: word frequency, bigrams, topic-wise word clouds and source comparison.

AI / ML task

NLP: Term Frequency-Inverse Document Frequency keywords, topic modeling or text clustering.
Optional classification: classify article source/topic from text.
Logistic Regression or SVM.

Minimum deliverables

Scraped dataset saved in SQLite/CSV.
Text cleaning pipeline and keyword table.
Comparative visualizations across two websites.
GitHub README explaining ethical scraping choices.

Evaluation and extension

Keyword quality: manual top-20 keyword review.
For classification: accuracy, macro-F1, confusion matrix.
Extension: named entity recognition or trend of keywords over time.

Suggested data / source links

1. 20 Newsgroups: https://scikit-learn.org/stable/modules/generated/sklearn.datasets.fetch_20newsgroups.html **2. News API:** https://newsapi.org/

6. Object Identification

Identify and count simple geometric shapes in photos or generated images.

Main idea

Detect/classify simple shapes: square, circle, triangle, pentagon and hexagon.
Try to learn computer vision at a beginner-friendly level.
Can be done using either image-processing rules or ML classification.

Data and EDA

Use generated geometric shape images or collect photos of printed shapes.
Keep background simple at first; then test on noisy/complex images.
EDA: number of images per class, image size, color/background variation.

AI / ML task

Classical CV: thresholding, contours, edges, shape features.
ML: KNN/SVM/Random Forest on shape features or CNN for advanced students.
Object count: detect contours and classify each detected object.

Minimum deliverables

Dataset folder with train/test split.
Notebook/script for preprocessing and feature extraction.
Classification result plus object-counting demo image.
GitHub with sample images and output screenshots.

Evaluation and extension

Metrics: accuracy, macro-F1 and confusion matrix.
For counting: compare predicted vs true count.
Extension: overlapping shapes, rotated shapes and multi-object scenes.

Suggested data / source links

1. Kaggle 2D shapes: https://www.kaggle.com/datasets/khalidboussaroual/2d-geometric-shapes-17-shapes **2. Kaggle simple shapes:** https://www.kaggle.com/datasets/dineshpiyasamara/geometric-shapes-dataset

7. Image Processing: Face Attendance

Recognize known faces in classroom images and write attendance to CSV.

Main idea

Build a face-recognition attendance prototype using known student images.
This is interesting but must be handled carefully because faces are personal data.
Recommended only with consent, small groups and local/offline processing.

Data and EDA

Collect 5-10 images per consenting student or use a public face dataset for practice.
Store only required labels; avoid sharing real student face data publicly.
EDA: image counts per person, lighting variation and failed detections.

AI / ML task

Face detection + embedding-based recognition.
Beginner route: use pretrained face-recognition embeddings, not train from scratch.
Attendance logic: if known face detected, append name/date/time to CSV.

Minimum deliverables

Consent/privacy note and data-handling rules.
Face-recognition notebook or app script.
Attendance CSV output and error cases.
GitHub code without uploading private student photos.

Evaluation and extension

Metrics: recognition accuracy, false accepted, false rejected.
Test under different light, distance and angle.
Extension: unknown-face detection and manual teacher verification.

Suggested data / source links

1. LFW faces: https://www.kaggle.com/datasets/jessicali9530/lfw-dataset **2. LFW people:** https://www.kaggle.com/datasets/atulanandjha/lfwpeople

8. Leaf / Feather Classification

Classify leaves or feathers from photos and include an “unknown” category.

Main idea

Take leaves of five trees or feathers of five birds and classify them from photos. Add one extra class for “unidentified/unknown”.

Data and EDA

Collect 40-100 photos per class if possible; use multiple angles/backgrounds.
Or use public plant-leaf/feather datasets for a cleaner starting point.
EDA: class counts, image size, color distribution and sample montage.

AI / ML task

Beginner: extract color/texture/shape features + Random Forest/SVM.
Intermediate: transfer learning using MobileNet/ResNet.
Unknown class: add out-of-distribution samples or confidence threshold.

Minimum deliverables

Dataset folder and metadata CSV.
Preprocessing pipeline: resize, normalize, train/test split.
Classification notebook with confusion matrix.
GitHub repository with sample images and model notes.

Evaluation and extension

Metrics: accuracy, macro-F1 and per-class recall.
Discuss which leaves/feathers are confused and why.
Extension: classify healthy vs unhealthy leaves or local tree species.

Suggested data / source links

1. Kaggle plant leaves: https://www.kaggle.com/datasets/csafrit2/plant-leaves-for-image-classification **2. Kaggle leaf classification:** https://www.kaggle.com/competitions/leaf-classification

9. Speech to Text

Convert English audio to text, then analyze or classify the transcribed text.

Main idea

Convert speech audio into text and save the output.
Use pretrained speech-to-text rather than training ASR from scratch.
Then perform text matching, keyword search or simple command classification.

Data and EDA

Use short English audio clips or public speech datasets.
Store audio filename, true transcript, predicted transcript and confidence/notes.
EDA: clip duration, word counts, error examples and transcript length.

AI / ML task

Use pretrained Automatic Speech Recognition model/library for transcription.
Text task: keyword detection, intent classification or match transcript to expected answer.
Advanced: compare two ASR systems or fine-tune a small model.

Minimum deliverables

Audio sample folder or public-data subset.
Speech-to-text script and transcript CSV/TXT files.
Text analysis or matching result.
GitHub README with setup and limitations.

Evaluation and extension

ASR metric: word error rate if true transcript is available.
Text task: accuracy/F1 or keyword precision/recall.
Extension: compare clear vs noisy audio or Indian-accented speech.

Suggested data / source links

1. LibriSpeech: https://www.openslr.org/12 **2. Mozilla Common Voice:** https://commonvoice.mozilla.org/en/datasets

10. Weather Data Analysis & Forecasting

Analyze temperature/rainfall patterns and predict next-day weather values.

Main idea

Analyze weather variables such as temperature, rainfall, humidity and wind.
Predict next-day temperature/rainfall or weather category.
Learn about time series, APIs and regression.

Data and EDA

Use weather APIs or public climate datasets.
Store daily city-wise data in SQLite/CSV.
EDA: monthly trends, seasonal effects, rainfall distribution and correlations.

AI / ML task

Regression: predict next-day temperature/rainfall.
Classification: rainy/not rainy or hot/normal/cold.
Models: baseline, Linear Regression, Decision Tree, Random Forest.

Minimum deliverables

API/data-download script and cleaned database.
EDA notebook with trend and seasonal plots.
Forecast model comparison.
GitHub repository with API-key safety instructions.

Evaluation and extension

Regression: MAE, RMSE, R^2.
Classification: accuracy, F1, confusion matrix.
Extension: compare predicted values with actual next-day weather. Also, use nearby cities data to improve prediction.

Suggested data / source links

1. WeatherAPI: https://www.weatherapi.com/ **2. OpenWeather:** https://openweathermap.org/api **3. NOAA NCDC:** https://www.ncdc.noaa.gov/cdo-web/datasets

11. Air Quality Prediction

Predict AQI or pollutant level using city, time and weather/pollutant features.

Main idea

Use Indian air-quality data to study PM2.5, PM10, NO2, SO2, CO, O3 and AQI.
Predict AQI value or classify air quality category.
Regression, classification and time-series EDA.

Data and EDA

Clean missing values, duplicate rows and inconsistent city/station names.
Create date features: month, season, weekday, festival/winter indicators.
Visualize city-wise AQI, pollutant correlations and yearly/monthly trends.

AI / ML task

Regression: Linear Regression, Decision Tree, Random Forest, XGBoost optional.
Classification: Good/Moderate/Poor/Very Poor/Severe categories.
Feature reduction: correlation filter, PCA or feature importance.

Minimum deliverables

SQLite database or CSV pipeline.
EDA notebook with pollutant and city comparison plots.
At least 3 models: baseline + 2 ML models.
GitHub repository with README and instructions.

Evaluation and extension

Regression metrics: MAE, RMSE, R^2.
Classification metrics: accuracy, precision, recall, F1, confusion matrix.
Extension: compare Delhi, Chandigarh, Mumbai or nearby cities; predict next-day AQI.

Suggested data links

1. Kaggle India AQI 2015-2020: https://www.kaggle.com/datasets/rohanrao/air-quality-data-in-india **2. Kaggle India AQI 2023-2025:** https://www.kaggle.com/datasets/saikiranudayana/india-air-quality-index-aqi-dataset-20232025

12. Student Performance Prediction

Predict marks or pass/fail category from study, family and school-related attributes.

Main idea

Predict final grade, score range or pass/fail status.
Be careful about ethics, privacy and responsible use of student data.

Data and EDA

Use anonymized public data; do not collect sensitive real student data without consent.
Features can include study time, attendance, previous marks, family support, activities.
EDA: grade distribution, correlations, effect of study time and absences.

AI / ML task

Regression: Linear Regression, Decision Tree, Random Forest.
Classification: Logistic Regression, Decision Tree, SVM.
Feature importance can show which variables are most predictive.

Minimum deliverables

Cleaned dataset and feature dictionary.
EDA notebook with 5-7 meaningful plots.
Baseline model plus 2 ML models.
Responsible AI note: avoid using model to punish students.

Evaluation and extension

Regression: MAE, RMSE, R^2.
Classification: accuracy, recall, F1, confusion matrix.
Extension: compare mathematics and literature datasets or create grade bands.

Suggested data links

1. UCI Student Performance: https://archive.ics.uci.edu/dataset/320/student **2. Kaggle Student Exams:** https://www.kaggle.com/datasets/spscientist/students-performance-in-exams

13. Music Genre Classification

Classify songs into genres using audio features rather than only listening.

Main idea

Use audio features such as MFCC, chroma, spectral centroid, tempo and energy.
Predict genre labels such as classical, rock, jazz, pop, disco or hip-hop.

Data and EDA

Start with extracted-feature CSV if audio processing is too hard.
Advanced one can use Librosa to extract features from .wav files.
EDA: compare tempo, spectral features and MFCC patterns across genres.

AI / ML task

Classification: KNN, SVM, Random Forest, Logistic Regression.
Feature scaling is important for KNN/SVM.
Optional deep learning: CNN on spectrogram images.

Minimum deliverables

Feature table with labels.
EDA notebook with genre comparison plots.
Model comparison and confusion matrix.
Short audio/AI reflection: which genres are confused and why?

Evaluation and extension

Metrics: accuracy, macro F1, confusion matrix.
Check class balance across genres.
Extension: build a simple song-similarity recommender using cosine similarity.

Suggested data links

1. Kaggle GTZAN Dataset: https://www.kaggle.com/datasets/andradaolteanu/gtzan-dataset-music-genre-classification **2. Kaggle Music Features:** https://www.kaggle.com/datasets/purumalgi/music-genre-classification

14. Indian Cricket Match Analysis

Analyze IPL data and predict match outcome or player performance patterns.

Main idea

Project using Indian Premier League data.
One can answer cricket questions using data instead of opinion.
Possible targets: match winner, total score range, player-of-match pattern.

Data and EDA

Use match-level data and also ball-by-ball data.
Features: venue, toss, batting first/second, team, season, overs, wickets.
EDA: toss impact, venue effect, team trends, batting first vs chasing.

AI / ML task

Classification: predict winner or high/low score category.
Regression: predict first-innings total or player runs.
Feature engineering: recent form, venue average, powerplay score.

Minimum deliverables

Clean match table and optional ball-by-ball aggregation.
EDA notebook with 6-8 cricket questions answered.
Model comparison with baseline.
Dashboard-style final summary or README.

Evaluation and extension

Classification: accuracy, F1, confusion matrix.
Regression: MAE/RMSE.
Extension: live-style prediction after 6 overs, 10 overs and 15 overs.

Suggested data links

1. Kaggle IPL Ball-by-Ball: https://www.kaggle.com/datasets/jamiewelsh2/ball-by-ball-ipl **2. Kaggle IPL 2008-2025:** https://www.kaggle.com/datasets/dgsports/ipl-ball-by-ball-2008-to-2022

15. House Price Prediction

Predict house price using area, location, rooms, amenities and property details.

Main idea

Classic beginner regression project with real-world meaning.
One do cleaning, categorical encoding and outlier handling.

Data and EDA

Features: area, bedrooms, bathrooms, location, furnishing, parking, year built.
EDA: price distribution, location effects, price per square foot, outliers.
Store cleaned dataset in SQLite or CSV.

AI / ML task

Regression: Linear Regression, Ridge/Lasso, Decision Tree, Random Forest.
Use one-hot encoding for categorical variables.
Feature reduction: correlation and feature importance.

Minimum deliverables

Data cleaning report with missing values and outliers.
EDA notebook with location and area analysis.
Three models: baseline mean/median + 2 ML models.
Prediction function for a new house.

Evaluation and extension

Metrics: MAE, RMSE, R^2.
Compare error in rupees and percentage error.
Extension: city-wise models or price-category classification.

Suggested data links

1. Kaggle India House Prices: https://www.kaggle.com/datasets/ankushpanday1/india-house-price-prediction **2. Kaggle House Prices Competition:** https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques

16. Fake News / News Category Classification

Classify news text as real/fake or assign topic categories using NLP.

Main idea

Introduces text data, NLP and responsible AI.
Safer beginning option: classify news category such as politics, sports, business, tech.
Fake-news detection is advanced/optional variant.

Data and EDA

Use headline + short description or full article text.
Clean text: lowercase, remove punctuation, stopwords, URLs and duplicates.
EDA: word counts, frequent terms, class balance and article length.

AI / ML task

Feature extraction: Bag of Words or TF-IDF.
Models: Naive Bayes, Logistic Regression, Linear SVM.
Optional: compare with pretrained embeddings for advanced students.

Minimum deliverables

Clean text pipeline and vectorizer.
EDA notebook with top words by category.
Confusion matrix and examples of misclassified headlines.
Responsible AI note: model predictions are not final truth.

Evaluation and extension

Metrics: accuracy, macro F1, precision, recall.
Check class imbalance and avoid overclaiming fake-news accuracy.
Extension: compare headline-only vs headline + description.

Suggested data links

1. Kaggle News Category: https://www.kaggle.com/datasets/rmisra/news-category-dataset **2. Kaggle WELFake News:** https://www.kaggle.com/datasets/saurabhshahane/fake-news-classification